

Article

Segmentation of Skin Lesions Using Deep YOLO-Family Networks: A Comparison of the Performance of Selected Models on a New Dataset

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Abstract

The aim of this study was to develop an effective and fast tool to support the automatic segmentation of skin lesions, with particular emphasis on the precise differentiation between malignant and benign lesions. In response to the problem of high false positive rates in existing CAD systems, modern neural network architectures from the YOLO family (YOLOv8, YOLOv9, YOLOv11, YOLOv12, and YOLOv26) were used in this research. The models were trained and evaluated on a new, balanced dataset (7000 images) based on the ISIC archive, where the key innovation was the introduction of a dedicated background class representing healthy skin. Through a multi-stage, rigorous optimization process, it was demonstrated that the yolov11s-seg model is highly effective for this task. It achieved a strong balance between effectiveness and processing speed, obtaining an mAP@50 score of 0.840 and an overall precision of 0.852. From a clinical perspective, the model's high sensitivity (85.9%) in detecting the most aggressive lesion, invasive melanoma (MI), is particularly noteworthy. Thanks to its extremely short inference time (only 4.8 ms), the proposed yolov11s-seg variant overcomes the limitations of heavy hybrid architecture, providing a stable and highly efficient solution showing significant potential for deployment in real-time medical mobile applications.

Keywords: instance segmentation; YOLOv11; healthy skin background; melanoma detection; dermoscopy; real-time mobile CAD



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1. Introduction

Skin cancer accounts for approximately one-third of all new cancer diagnoses worldwide [1,2]. Although melanoma accounts for only 1–4% of cases, it is responsible for approximately 80% of all skin cancer-related deaths [3–5]. It is often considered the most dangerous type of skin cancer and ranks 23rd among cancers with the highest mortality rate [1]. Importantly, early detection and treatment significantly increase the chances of survival [6,7]. Detecting melanoma at the first or second stage predicts a five-year survival rate of 97–99%. The survival rate drops drastically to 32% for the third stage [8]. At the fourth stage, a melanoma diagnosis may leave the patient with only 0.6 to 1.4 years of

life [9]. Early detection not only improves the prognosis but also reduces the costs and invasiveness of treatment [10].

Only about 60% of skin cancer cases are correctly diagnosed based solely on visual examinations [11]. The similarity between melanoma and benign skin lesions, such as dermatofibroma or seborrheic keratosis, leads to many excessive biopsies, resulting in unnecessary costs and scarring [12,13]. Due to the high risk associated with melanoma, there are 15 to 30 mistaken biopsies of benign skin lesions for every melanoma biopsy. A biopsy is invasive, painful, leaves scars, and carries a risk of infection [14]. Given these facts, it can be concluded that manual diagnosis suffers from several problems, such as overdiagnosis and consequent unnecessary biopsies, driven by the fear of missing a melanoma [15]. Diagnostic accuracy largely depends on the physician's visual acuity and experience in classifying subtle structures [16]. Pathologists disagree with each other in about 8% to 26% of cases [17]. These differences in interpretation can lead to both the overtreatment of benign lesions and the undertreatment of malignant ones [18].

Therefore, computer-aided diagnosis (CAD) systems are implemented to improve diagnostic accuracy and reduce both the overtreatment of benign lesions and the undertreatment of malignant ones. A key first step in the operation of CAD systems is segmentation, which involves isolating the lesion from the background [19–22]. This task is challenging due to the presence of artifacts such as hair, dermoscopy gel, gel bubbles, rulers, noise, and blood vessels. The color similarity between the lesions and the surrounding skin poses an additional problem. Consequently, there is a demand for automated solutions capable of detecting lesion boundaries despite these obstacles [23–28]. Precise lesion segmentation can enhance classification accuracy, transparency, reliability, and model explainability by clearly defining a specific area as its focal point. This provides a method for validating the outcomes of the model's decision-making process [29].

In systems used in clinical settings, there is a need for high accuracy and speed with minimal delay. The systems should be capable of processing information and returning results in real time, while remaining computationally efficient enough to operate on mobile and energy-efficient devices [30]. For this reason, models based on the YOLO (You Only Look Once) architecture are an appropriate choice for segmentation and classification tasks in real-time clinical environments [31–36]. YOLO models offer a balance between speed, efficiency, and accuracy [30,34]. This is due to their low computational overhead and small size, which make them ideal for deployment on devices with limited hardware resources [33,37].

The main contribution of this work to the field of automated dermatological diagnosis encompasses six key aspects:

- A new, realistic dataset. This work introduces a new, balanced dataset (7000 images) based on the ISIC archive, which—unlike many existing datasets—preserves the natural variability in image quality, artifacts, and resolution diversity. This approach increases the realism of experiments and their practical value, which is crucial in clinical applications.
- Introduction of a background class (healthy skin). An important and original contribution of this work is the extension of the dataset with a healthy skin class. This enables models to learn to distinguish pathological lesions from unaffected areas, directly addressing the problem of high false positive rates—one of the main limitations of existing CAD systems.
- Publicly available, fully annotated dataset. The created dataset, along with detailed annotations in segmentation format (polygons), has been made publicly available. This ensures full reproducibility of the research and constitutes a tangible contribution to the scientific community, enabling further comparisons and method development.

To the authors' knowledge, there is currently no publicly available dataset of this size with such detailed annotations for segmentation purposes.

- Systematic and up-to-date analysis of YOLO models for segmentation. The work presents one of the first such comprehensive analyses of the latest models from the YOLO family (including variants such as yolov12 and yolov26) in the context of skin lesion segmentation. The contribution lies in providing structured knowledge about their real capabilities, trade-offs, and clinical applicability.
- Multi-criteria model evaluation methodology. A structured, multi-stage method for model selection is proposed, simultaneously considering:
 - recall (crucial in medical diagnostics),
 - localization accuracy (mAP@50–95),
 - inference time.

This methodology is general in nature and can be applied to other tasks in medical image processing.

- Deployment-oriented approach—mobile system architecture. Unlike many purely theoretical studies, this work also presents a complete concept of a client–server system architecture for mobile applications, considering real computational constraints and application requirements.

This paper presents the process of developing an autonomous skin lesion recognition system, whose primary objective is to enable the differentiation between benign nevi and malignant melanoma. The study compares five YOLO architecture models (YOLOv8, YOLOv9, YOLOv11, YOLOv12, and YOLOv26). The main goal of this comparison was to identify the model that maintains the best possible balance between accuracy and inference speed, making it optimal for deployment in real-world clinical conditions. To clearly present the research, the article is organized into the following sections: Section 2 outlines the contributions of other researchers in the field of medical imaging and segmentation using YOLO models. Section 3 discusses the dataset structure, presents the data acquisition and preparation process, and justifies the decisions made during this stage. Section 4 details the training and evaluation processes used to select the optimal model. Finally, Section 5 compares the obtained results with the findings of other authors and proposes a system for the practical application of these results.

2. Related Works

YOLO family models have gained significant popularity in medical image analysis due to their favorable speed-to-accuracy ratio. Early works by Aishwarya et al. [38] demonstrated the effectiveness of the YOLOv3 and YOLOv4 variants in skin lesion detection, where the YOLOv4 model achieved an mAP@50 score of 0.865, Precision of 0.92, Recall of 0.91, and F1-score of 0.92. Newer versions of the architecture have brought further improvements to these metrics; İnan et al. [39] demonstrated the superiority of the YOLOv11 model over YOLOv8 (with 22% fewer parameters), achieving a Precision of 0.92, Recall of 0.98, F1-score of 0.95, and mAP@50 of 0.92 for YOLOv11. The advantage of the latest architectures over classic CNNs (such as ResNet50 or VGG16) was also confirmed by Diphtho and Basak [40], where the YOLOv11 model achieved a Precision of 0.887, Recall of 0.867, and F1-score of 0.87 in classification tasks. YOLO architectures are also applied directly to pixel-level segmentation. Albahli [31] used the YOLOv8 model for this task, reporting excellent results: an mAP@50 score of 0.986, Precision of 0.91, Recall of 0.90, Dice coefficient of 0.92, and IoU of 0.88, with an inference time of only 12.5 ms (80 FPS).

A contemporary trend for handling difficult artifacts (such as body hair and lack of contrast) is the development of hybrid architectures. Gül et al. [30] proposed the

“YOLOSAMIC” approach, combining YOLOv8 detection with Segment Anything Model (SAM) segmentation, achieving a Dice coefficient of 0.939, IoU of 0.911, Precision of 0.979, Recall of 0.930, and mAP@50 of 0.983 on public datasets. A similar hybrid combining YOLOv10 and SAM was applied by Nagaoka [41], who reported a mean IoU of 0.738, Precision of 0.954, Recall of 0.772, and F1-score of 0.833. Hybrid models also include combining fast single-stage detectors with precise two-stage ones; Marie et al. [42] integrated YOLOv9 with Faster R-CNN, achieving an mAP@50-95 of 0.895, F1-score of 0.951, Precision of 0.951, and sensitivity of 0.944 (at a speed of 31.3 FPS). Furthermore, Mio et al. [43] used a two-stage DBB-YOLO and UNet algorithm for melanonychia segmentation, where the detection stage achieved an F1-score of 0.986, and the segmentation stage reached a Dice coefficient of 0.728 and Recall of 0.795. On the other hand, in studies focused on diagnostic explainability, AbuAlkebash et al. [44] combined YOLOv8 with Vision Transformers (ViT). This solution yielded an overall mAP@50 of 0.87, Precision of 0.742, and Recall of 0.717; however, it suffered from a noticeable drop in effectiveness in isolated melanoma classification (Recall of only 0.609).

Despite their dynamic development and the high results presented in the literature, these models still encounter problems in highly noisy, real-world clinical conditions. Existing studies still lack rigorous analyses that systematically utilize datasets supplemented with a healthy skin background (to reduce false-positive detections). Furthermore, there is a lack of a comprehensive comparison regarding the capabilities of the latest innovations—including the YOLOv12 and YOLOv26 architectures—in the instance segmentation task for dermoscopic images.

3. Materials and Methods

3.1. Dataset

The research material was collected using the ISIC archive, which currently offers the largest volume and variety of dermoscopic images. The following types of skin lesions were selected from this dataset:

- Malignant melanoma (invasive)—the most aggressive skin cancer, with a high risk of metastasis and life threat. It accounts for 3–5% of skin cancers but is responsible for most deaths caused by this type of malignancy. The ISIC archive contains images of several melanoma categories. Invasive melanoma, the most dangerous category for the patient capable of metastasizing, was used in this study.
- Basal cell carcinoma—the most common skin cancer, accounting for approximately 80% of all malignant skin neoplasms. It is low-grade and rarely metastasizes, but it can destroy surrounding tissues, such as bones (it is locally malignant).
- Squamous cell carcinoma—the second most common skin cancer, accounting for about 15–20% of malignant skin neoplasms. It is also locally malignant, like basal cell carcinoma, but differs by being more aggressive.
- Melanocytic nevus—a benign lesion occurring in over 90% of adults. In Caucasians, an average person has about 20–40 such lesions. Atypical nevi are much rarer (about 2–10% of the population) but can transform into melanoma.
- Pigmented benign keratosis—a very common benign lesion that is often confused with melanoma or other skin cancers. It occurs in most people over the age of forty or fifty.

As can be seen from the above list, the selection of skin lesions for this study was guided by their level of aggressiveness (malignant melanoma) and their frequency of occurrence—basal cell and squamous cell carcinomas for malignant lesions, and melanocytic nevus and pigmented benign keratosis for benign lesions. In addition to skin lesions, a certain number of background images presenting healthy skin without any

lesions were also collected. The use of background images aimed to reduce the number of false-positive detections. Example images used in the research are presented in Figure 1.

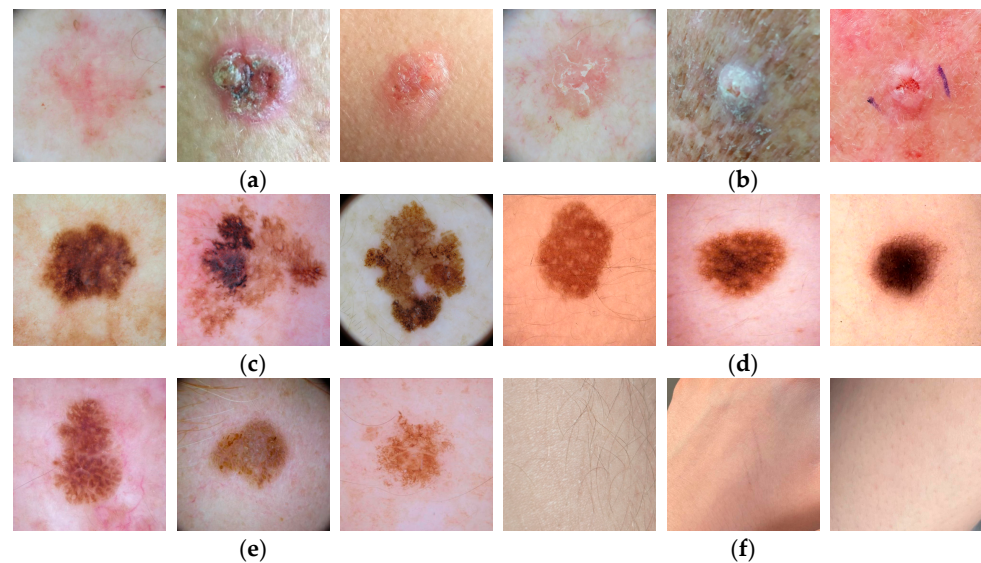


Figure 1. Examples of skin lesions: (a) basal cell carcinoma; (b) squamous cell carcinoma; (c) melanoma invasive; (d) nevus; (e) pigmented benign keratosis; (f) background images. To facilitate observation, the lesions were cropped with an aspect ratio of 1:1.

3.2. Dataset Preparation

From the downloaded dataset, the following numbers of images belonging to the individual observation classes were randomly selected for further research: melanoma invasive (MI)—1000, basal cell carcinoma (BCC)—1000, squamous cell carcinoma (SCC)—1000, melanocytic nevus (N)—500, pigmented benign keratosis (PBK)—500. The number of background images collected (BGD) was 1000. The preliminary filtering of the dataset was intentionally not performed, resulting in images with a wide range of resolutions, artifact content, contrast, and visibility levels of the regions of interest against the skin background; at the same time, the dataset is balanced across all classes, including the background (healthy skin), and deliberately preserves real-world variability—such as hair occlusions, illumination differences, noise, and blur—to reflect realistic dermoscopic imaging conditions. Thanks to the greater realism of the images used in the study, the results obtained are more reliable.

Preliminary studies have shown that basal cell and squamous cell carcinomas are detected with the lowest sensitivity. In this case, there is also a high number of errors resulting from the misidentification of the first cancer as the second and vice versa. This situation results from the fact that both cancers are visually very similar to each other and usually stand out very poorly against the skin (Figure 1a,b). Therefore, we combined both cancer categories into a single class designated as BCC+SCC. In our opinion, such a decision will not have negative consequences for the patient, because upon a positive diagnosis, the patient will be referred to more detailed dermatological diagnostics.

Image annotation in the YOLO format was performed using the online Supervisely platform. This process was carried out after consultation with dermatologists from the University Clinical Hospital No. 1 (Lublin, Poland). After annotating the images, the full dataset was randomly divided into training, validation, and test sets such that the number of images for each class (and background images) in the validation and test sets was 100. In the case of the training set, the number of images in individual classes was as follows: BCC+SCC—1800, MI—800, N—300, PBK—300, BGD—800. To balance the number of samples in individual classes, augmentation of the images from the MI, N, and PBK classes

belonging solely to the training set was performed. The Albumentations 2.0.8 package was used for this purpose. To transform the images, combinations of the following functions were applied:

- HorizontalFlip()—flips the input left to right (around the y -axis).
- SafeRotate()—rotates the input inside the input's frame by an angle selected randomly from the uniform distribution.
- RandomBrightnessContrast()—randomly changes the brightness and contrast of the input image.

After performing augmentation, the total number of images amounted to 7000. Figure 2 provides detailed information on the number of skin lesions belonging to individual classes (and background images) for each of the subsets used in the study. The prepared dataset was published on Zenodo platform (<https://zenodo.org/records/18938292>, accessed on 10 March 2026).

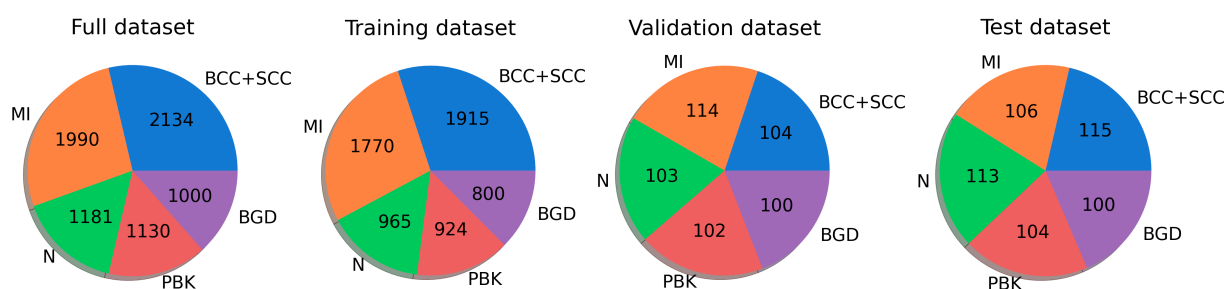


Figure 2. The structure of the datasets used. The pie charts provide information on the number of skin lesions belonging to a specific class.

The process flow of the conducted research is presented in Figure 3. The training and validation sets were used to build 22 models for skin lesion segmentation. Next, the models were evaluated using the test set. Based on the evaluation results, the analysis and selection of the best model were performed. In the final stage, the selected model was used for sample prediction of images that had not been used for training and validation.

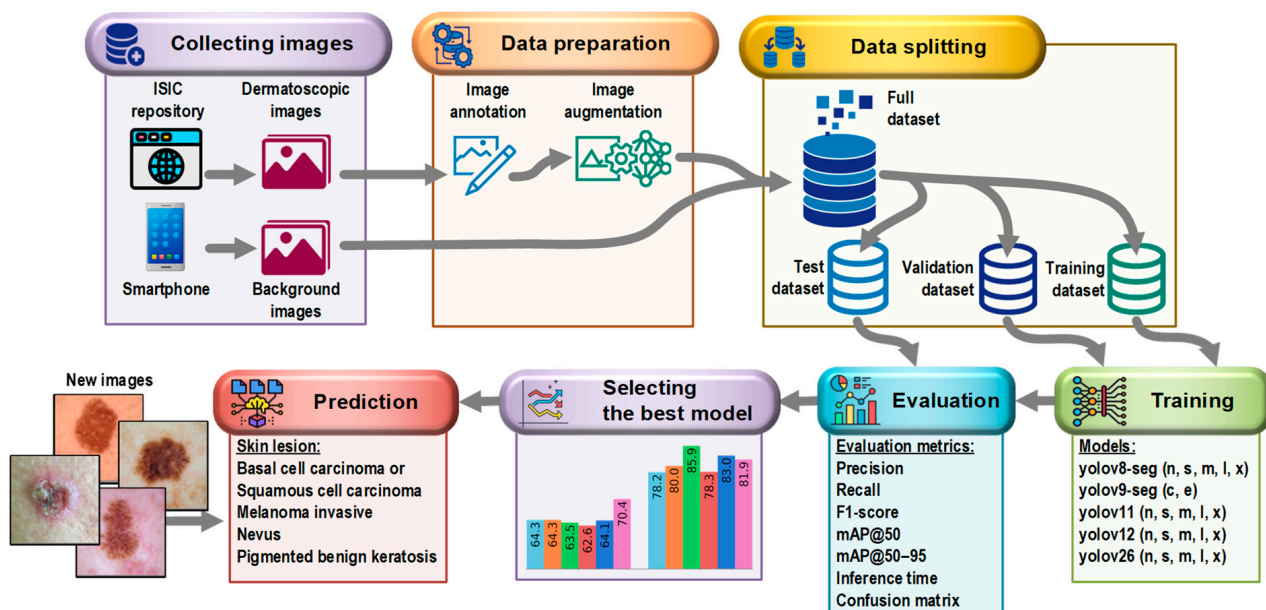


Figure 3. The processes flow of the research performed.

3.3. Metrics

In this study, the segmentation quality of the YOLO models was assessed with the use of the following metrics: inference time, precision, recall, F1-score, mAP@50, and mAP@50-95, aiming to provide a multi-dimensional outlook of the performance of each of the models.

A True Positive (TP) constitutes a correct detection of the ground-truth pixels. A False Positive (FP) can be defined as an incorrect detection of a nonexistent pixel. A False Negative (FN) represents a case of the ground-truth pixel remaining undetected.

Precision evaluates the model's ability to correctly detect relevant objects (1). It measures the percentage of accurate positive predictions to all the positive predictions. Precision is mathematically defined by the ratio of True Positives to all detections:

$$Precision = \frac{TP}{TP + FP} = \frac{TP}{all\ detections} \quad (1)$$

Recall evaluates the model's ability to correctly identify all relevant cases (2). It describes the "completeness" of our positive predictions relative to reality and answers the question of how many pixels we managed to detect out of all those that were marked using reference masks. Recall is mathematically defined as the ratio of true positive pixels to the total ground-truth pixels:

$$Recall = \frac{TP}{TP + FN} = \frac{TP}{all\ ground - truths} \quad (2)$$

The F1-score can be defined as the harmonic mean of precision and recall (3). It is calculated using the formula:

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (3)$$

The mAP (mean Average Precision) is a metric used for the evaluation of object detector performance across all classes in a dataset (4). The mAP is calculated as the average of the Average Precision (AP) over all evaluated dataset classes with the formula:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (4)$$

N is the total number of classes and AP is the AP in the class.

The mAP must be evaluated at a specific intersection over Union (IoU) threshold.

The mAP@50 measures the Average Precision using a single IoU threshold set at 50%.

The mAP@50-95 averages the Average Precision over 10 different IoU thresholds varying in range from 50% to 95%, increasing in 5% steps [45].

3.4. The YOLO Family of Models

Although the YOLO family of models has undergone significant evolution since its inception, this study focused on evaluating the latest-generation architectures: YOLOv8, YOLOv9, YOLOv11, YOLOv12, and YOLOv26. This choice was dictated by the desire to compare models with a widely recognized position and stability (YOLOv8) against innovative iterations emphasizing attention mechanisms (YOLOv11, YOLOv12) and extreme optimization for edge devices (YOLOv26). Older versions of the algorithm were omitted from the detailed analysis because, in terms of segmentation precision, they have been significantly marginalized in favor of newer solutions. A summary of the key innovations for each architecture is presented in Table 1.

Table 1. Comparison of features of chosen YOLO architectures.

Model	Key Innovations and Architectural Features	Significance in the Context of Our Research
YOLOv8	Anchor-free architecture, decoupled head, replacement of the standard layer with the C2f module.	Provides a stable, highly efficient reference point for the trade-off between accuracy and speed.
YOLOv9	Introduction of PGI (Programmable Gradient Information) and the GLEAN architecture, minimizing data loss.	Preventing feature degradation in deep layers supports the segmentation of poorly defined skin lesion boundaries.
YOLOv11	Use of C3k2 blocks and the C2PSA (Cross Stage Partial Attention) spatial attention mechanism.	The attention mechanism allows the network to focus on the most relevant and minute fragments of the lesion.
YOLOv12	Attention-centric architecture (e.g., FlashAttention), R-ELAN residual connections, separable convolutions.	Verification of whether such advanced attention mechanisms translate into improved medical diagnostic efficacy.
YOLOv26	Architecture optimized for edge computing, absence of NMS (Non-Maximum Suppression) in the post-processing phase, improved STAL mechanism.	Evaluation of the model's suitability for smooth, real-time operation on mobile devices with limited resources.

In each of the examined cases, specific model variants designed for instance segmentation tasks (denoted by the “-seg” suffix) were used, offered in various weight sizes (from nano to extra-large). Segmentation based on the YOLO architecture uses a dedicated head branch to generate prototype masks. This allows not only for instantaneous object detection but, above all, to determine its precise boundaries at the pixel level. Such functionality is a critical requirement in automating the calculation of the ABCDE score and in building reliable computer-aided diagnosis (CAD) systems.

4. Results

4.1. Training and Evaluation Process

In this study, we employed the following network architectures capable of solving the object segmentation task: YOLOv8, YOLOv9, YOLOv11, YOLOv12, and YOLOv26. The choice of a given architecture depended on whether it allowed fine-tuning the model weights to our own dataset. The individual model variants belonging to these architectures were pre-trained on the COCO dataset, which contains 80 object classes. Ultralytics provides an API that supports the model training and testing process. Thanks to this and the transfer learning technique, the pre-trained model weights were fine-tuned to 4 categories of skin lesions. All models were trained using the default YOLO-family hyperparameters, which are widely validated and provide stable convergence. These settings ensure robust generalization on our diverse dataset and prevent overfitting. Default loss weights and NMS thresholds maintain a balanced optimization between localization, classification, and segmentation. Additionally, these parameters support fast inference, enabling real-time deployment in mobile medical applications. Table 2 contains the description and values of selected hyperparameters used during the model training process.

The construction of the models was carried out on the NVIDIA DGX A100 computing platform, the main components of which were as follows: CPU—Dual AMD Rome 7742, 256 cores, 1 TB; GPU—8 × NVIDIA A100 SXM4 80 GB Tensor Core; OS: Ubuntu 22.04.5 LTS. The torch 2.3.0 platform and Python 3.10.12 programming language were used to build the models. Upon completion of the training process, all models were evaluated using the test set. The results of this process are presented in Table 3. As a result of the evaluation, object segmentation models return two groups of quality metrics. The first group is based on the prediction of the object's bounding box, while the second group is based on the

prediction of the object's mask. In Table 3 and the subsequent analysis, we exclusively used the results obtained based on mask prediction. This was because we were investigating models designed for segmentation rather than object detection; therefore, object properties based on mask segmentation were of greater interest to us.

Table 2. Selected hyperparameters used during the models training process.

Description	Name and Value
Total number of training epochs	epochs = 50
Number of epochs to wait without improvement in validation metrics before early stopping the training	patience = 10
Batch size	batch = 16
Target image size for training	imgsz = 640
Choice of optimizer for training	optimizer = SGD
Momentum factor influencing the incorporation of past gradients in the current update	momentum = 0.937
Initial learning rate	lr0 = 0.01
Final learning rate as a fraction of the initial rate ($lr0 \times lrf$), used in conjunction with schedulers to adjust the learning rate over time	lrf = 0.01
L2 regularization term, penalizing large weights to prevent overfitting	weight_decay = 0.0005
Number of epochs for learning rate warmup, gradually increasing the learning rate from a low value to the initial learning rate to stabilize training early on	warmup_epochs = 3.0
Initial momentum for warmup phase, gradually adjusting to the set momentum over the warmup period	warmup_momentum = 0.8
Learning rate for bias parameters during the warmup phase, helping stabilize model training in the initial epochs	warmup_bias_lr = 0.1
Weight of the box loss component in the loss function, influencing how much emphasis is placed on accurately predicting bounding box coordinates	box = 7.5
Weight of the classification loss in the total loss function, affecting the importance of correct class prediction relative to other components	cls = 0.5
Weight of the distribution focal loss, used in certain YOLO versions for fine-grained classification	dfl = 1.5
Sets the Intersection Over Union threshold for Non-Maximum Suppression	iou = 0.7
Limits the maximum number of detections per image	max_det = 300
Enables test-time augmentation during validation	augment = false

Analyzing the results obtained in Table 3, a significant impact of the model's depth on its performance in the segmentation task was observed. On the one hand, extensive models such as yolov26x dominated in terms of overall precision (0.878) and the mAP@50-95 metric (0.624); however, their long inference time of 817.5 ms disqualifies them from smooth operation on average consumer devices. On the other hand, models designated with the 's' and 'l' suffixes were characterized by the best stability in sensitivity (Recall), achieving values ranging from 0.756 (yolov8s) to as high as 0.805 in the fastest yolov11s model (4.8 ms). The results clearly suggest that the 'heaviest' versions overfit to complex edges at the expense of overall sensitivity, which is why 'lighter architectures' are much more effective for diagnostic purposes in this case.

Table 3. Performance metrics for all tested models.

Architecture	Version	Parameters (M) /GFLOPs	Precision	Recall	F1-Score	mAP@50	mAP@50–95	Inf. Time (ms)
YOLOv8	n	3.3/11.3	0.82	0.753	0.785	0.809	0.581	3.1
	s	11.8/39.9	0.793	0.756	0.774	0.804	0.575	4.7
	m	27.2/104.3	0.795	0.77	0.782	0.81	0.583	9.2
	l	45.9/210.1	0.836	0.777	0.805	0.827	0.602	14.2
	x	71.7/328.0	0.827	0.779	0.802	0.804	0.581	23
YOLOv9	c	27.6/147.6	0.838	0.776	0.806	0.828	0.6	14
	e	59.7/234.4	0.773	0.767	0.77	0.803	0.591	15.1
YOLOv11	n	2.8/9.6	0.841	0.767	0.802	0.824	0.601	2.6
	s	10.1/32.8	0.823	0.805	0.814	0.84	0.621	4.8
	m	22.3/112.9	0.821	0.771	0.796	0.824	0.603	10.5
	l	27.6/131.8	0.852	0.779	0.814	0.835	0.614	12.9
	x	62.0/295.9	0.782	0.759	0.77	0.798	0.593	18.7
YOLOv12	n	2.8/9.1	0.821	0.769	0.794	0.816	0.585	3.5
	s	9.8/30.8	0.75	0.788	0.768	0.802	0.579	6.4
	m	21.9/104.8	0.79	0.759	0.774	0.796	0.583	13.8
	l	28.2/126.3	0.8	0.756	0.778	0.801	0.585	19.1
	x	63.1/283.0	0.823	0.755	0.788	0.81	0.589	35.9
YOLOv26	n	2.7/9.0	0.808	0.745	0.775	0.801	0.58	52.1
	s	10.4/34.1	0.808	0.763	0.785	0.803	0.581	149.4
	m	23.5/121.2	0.841	0.732	0.783	0.793	0.578	387.8
	l	27.9/139.4	0.821	0.804	0.812	0.821	0.607	449
	x	62.7/313.0	0.878	0.773	0.822	0.837	0.624	817.5

4.2. Selecting the Best Model

The selection of the best model was conducted using a two-stage analysis. In the first stage, comprehensive quality metrics were applied in the form of the F1-score—evaluating the trade-off between precision and sensitivity (Recall)—and the mAP@50-95 metric, which averages mAP values across IoU thresholds ranging from 0.50 to 0.95. The second stage focused on metrics combining the models’ effectiveness and efficiency. Here, sensitivity (Recall)—which is crucial in medical applications—and inference time, which allows for assessing the feasibility of practical model implementation under specific hardware and software conditions, were utilized.

It is convenient to compare the comprehensive effectiveness of the models based on Figure 4. The points in this figure represent pairs of values (mAP@50-95, F1-score). Assuming that the values of both metrics should be as large as possible, we expect the points representing the models of interest to be in the upper-right area of the chart. The symbols of the models meeting our expectations have been enclosed by a dashed gray line. Based on the above-presented effectiveness comparison, the following models were included in the further analysis: yolov8l-seg, yolov9c-seg, yolov11n-seg, yolov11s-seg, yolov11m-seg, yolov11l-seg, yolov26l-seg, and yolov26x-seg.

The values of the basic quality metrics for the models selected based on Figure 4 are presented in the form of bar charts in Figure 5. These charts display global metric values, encompassing all observation classes collectively. It should be emphasized that the highest sensitivity values were achieved by the yolov11s-seg (80.5%) and yolov26l-seg (80.4%) models. For the second most important metric, mAP@50-95, the yolov26x-seg (62.4%) and, again, the yolov11s-seg (62.1%) models proved to be the best. Note that the yolov26x-seg model also achieved the highest precision value (87.8%), ahead of the second-ranked yolov11l-seg model (85.2%). The fourth metric is mAP@50. In this case, the yolov11s-seg achieved the highest score (84.0%), slightly ahead of the yolov26x-seg model (83.7%).

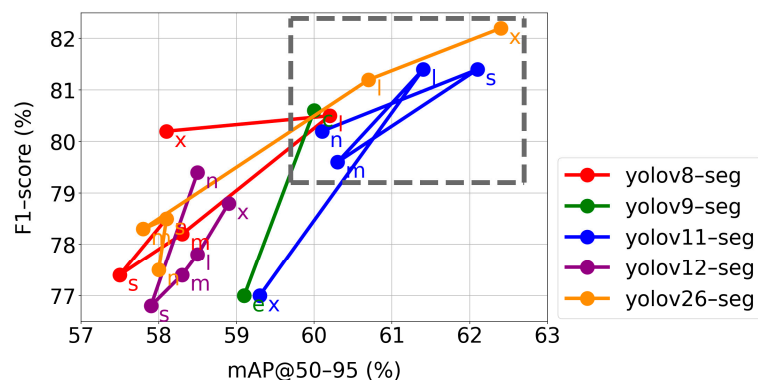


Figure 4. F1-score and mAP@50–95 values of the models built in this study for all classes together.

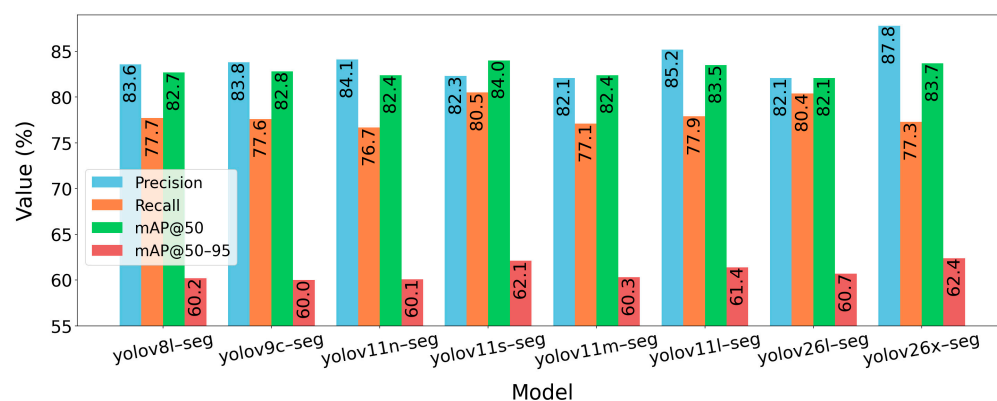


Figure 5. Global evaluation metrics of models selected from Figure 4.

The second stage of the analysis considered the models' sensitivity and inference time. The points representing pairs of these values are shown in Figure 6. In this case, we expect the model symbols to be in the upper-left area of the chart. In Figure 6, a dashed gray line encloses the symbols of those models that, on the one hand, meet the previous expectation, and on the other, were selected based on Figure 4. Among the models selected at this stage were: yolov8l-seg, yolov9c-seg, yolov11s-seg, yolov11l-seg, and yolov12s-seg. Additionally, the yolov26l-seg model was also selected. Although it was characterized by a very long inference time (449 ms), its sensitivity (80.4%) was comparable to the highest sensitivity of the yolov11s-seg model (80.5%). Therefore, we did not want to eliminate this model from further analysis.

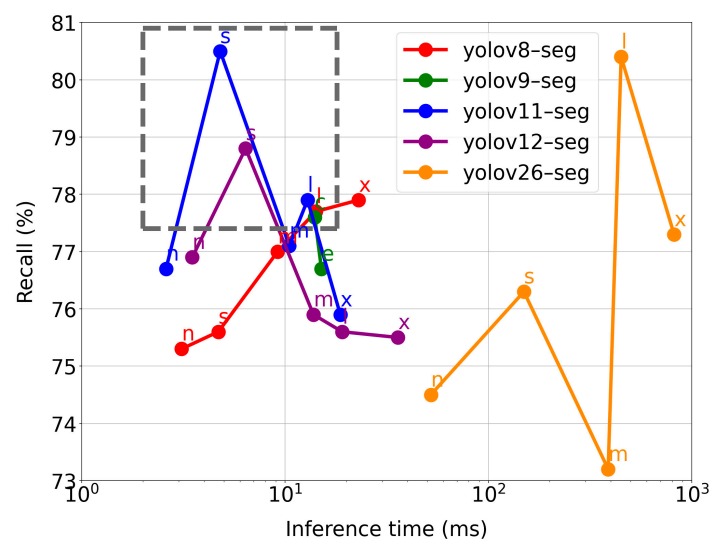


Figure 6. Recall and inference time values of the models built in this study for all classes together.

The sensitivity of the models selected based on Figure 6 is presented in the form of bar charts in Figure 7. These charts illustrate the sensitivity obtained for individual skin lesions. The most interesting results concern malignant lesions. From a medical standpoint, high sensitivity in detecting these lesions is beneficial for patients. In the case of the MI class, the yolov11s-seg model achieved the highest sensitivity (85.9%), followed by the yolov12s-seg model in second place (83.0%). The cases that proved the most difficult to detect for all models belonged to the BCC+SCC class. By far, the highest effectiveness in this case was demonstrated by the yolov26l-seg model. It achieved a sensitivity of 70.4%, which turned out to be over 6% higher than the sensitivity of the other models. For the two benign lesions, significantly better results were obtained for the PBK class. The most effective in this instance proved to be the yolov11s-seg model, with a sensitivity of 94.2%. The results obtained for the N class were markedly worse. The sensitivity of the best model, yolov26l-seg, was 78.8% in this case. Comparing the results obtained for individual skin lesions, it should be noted that two models proved particularly effective. The yolov11s-seg model achieved the highest sensitivity for the MI (85.9%) and PBK (94.2%) classes. Meanwhile, the sensitivity of the yolov26l-seg model was the highest for the BCC+SCC (70.4%) and N (78.8%) classes.

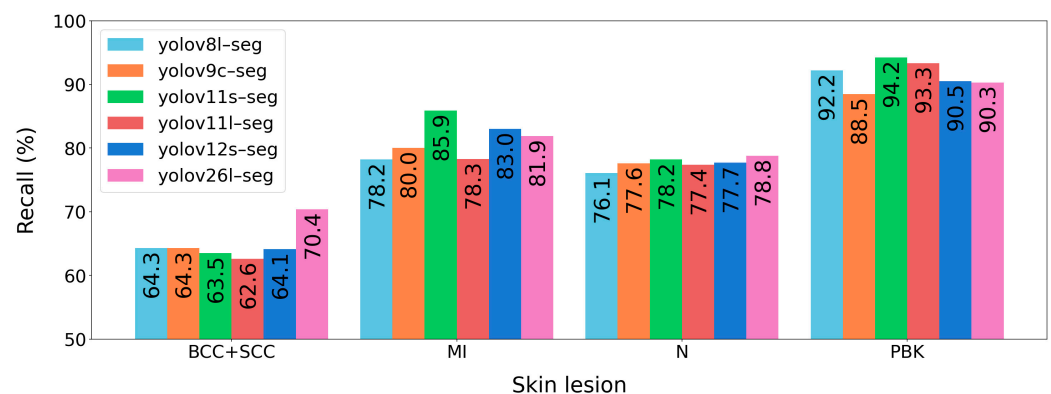


Figure 7. Recall of detection of individual skin lesions by models selected from Figure 6.

As is evident from the earlier comparison, the yolov11s-seg and yolov26l-seg models clearly stand out in terms of effectiveness. However, considering the patient's well-being, we selected the first of them, the yolov11s-seg, as the best model. This is because this model has a significantly higher sensitivity in detecting the most malignant skin lesion, which is invasive malignant melanoma (MI class).

4.3. Results of the Best Model

Based on the conducted two-stage verification, the yolov11s-seg model was considered the best solution. Analyzing the detailed metrics in Figure 7, we can see that its remarkable effectiveness lies in an extremely high sensitivity in identifying key lesions: invasive melanoma (MI class—85.9%) and pigmented benign keratosis (PBK—94.2%). A detailed evaluation of the yolov11s-seg model's effectiveness on the test set is illustrated by the confusion matrix (Figure 8). An analysis of the confusion matrix reveals that the most serious challenge for the network was the correct classification and segmentation of BCC and SCC tumors (the combined class). This is a natural appearance, considering that these lesions are blurred and stand out poorly against the background, often being masked by elements such as body hair.

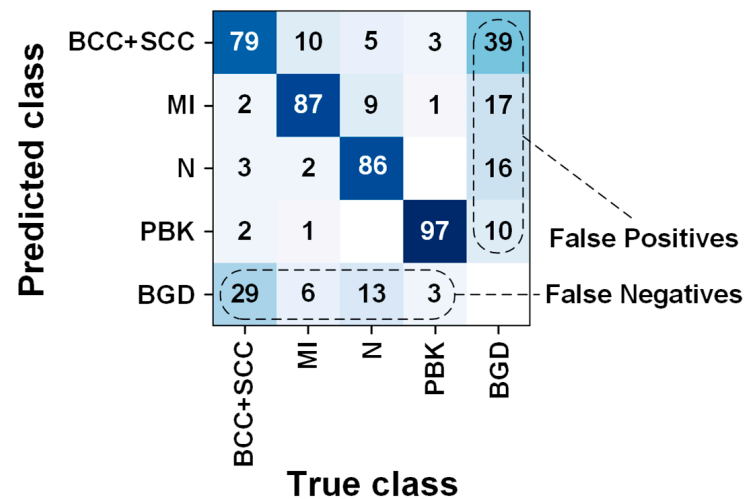


Figure 8. The yolov11s-seg model confusion matrix built for the test set.

The dynamics and stability of the algorithm's learning process are presented in Figure 9. It contains the plots of training and validation losses (Figure 9a), as well as the progression of basic segmentation quality metrics on the validation set (Figure 9b) over successive epochs. In turn, a comprehensive evaluation of the model's decision-making capabilities for the test set is presented in Figure 10. These plots illustrate the key relationships between the metrics and the confidence threshold, including the following curves: precision–confidence (Figure 10a), recall–confidence (Figure 10b), and F1 score–confidence (Figure 10c), as well as the classic precision–recall curve (Figure 10d).

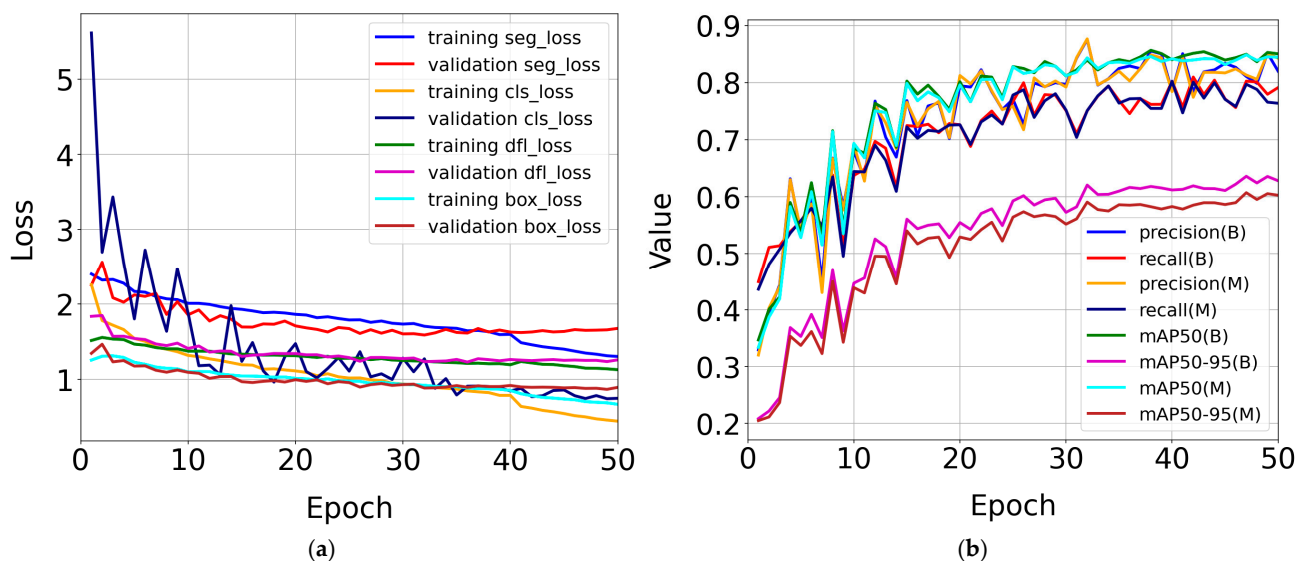


Figure 9. Plots of training and validation losses (a) and plots of basic segmentation quality metrics for the validation set (b) built for the yolov11s-seg model.

From a practical perspective, the effectiveness of the algorithm is demonstrated by the prediction examples in Figure 11, which compares the results for lesions identified with higher (Figure 11a) and lower (Figure 11b) confidence levels. The red line indicates the boundaries of the reference mask (ground-truth) on them. It can be observed that despite the lack of perfect contrast, in the case of lesions identified with a higher confidence level, the model correctly traces the segmentation mask along the designated edges.

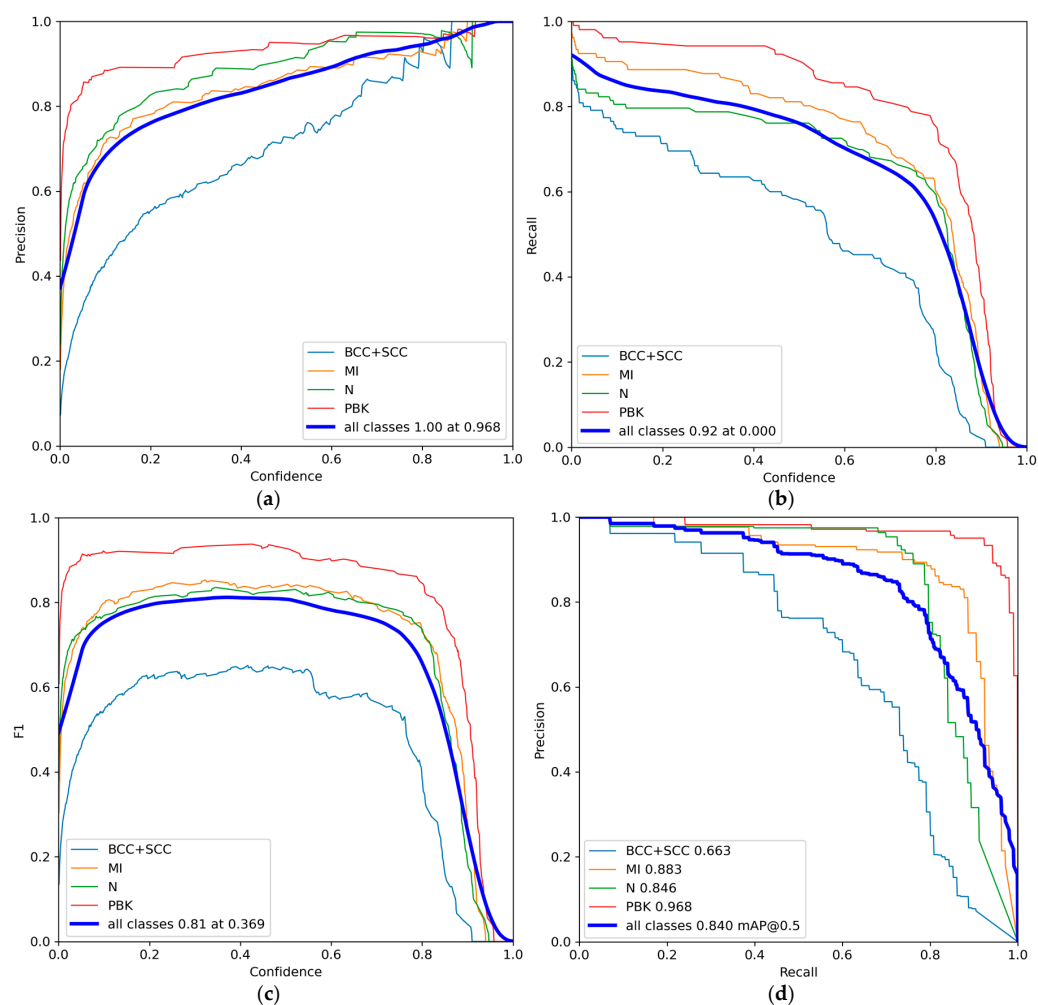


Figure 10. Plots of the yolov11s-seg model segmentation quality metrics for the test set: (a) precision–confidence; (b) recall–confidence; (c) F1 score–confidence; (d) precision–recall.

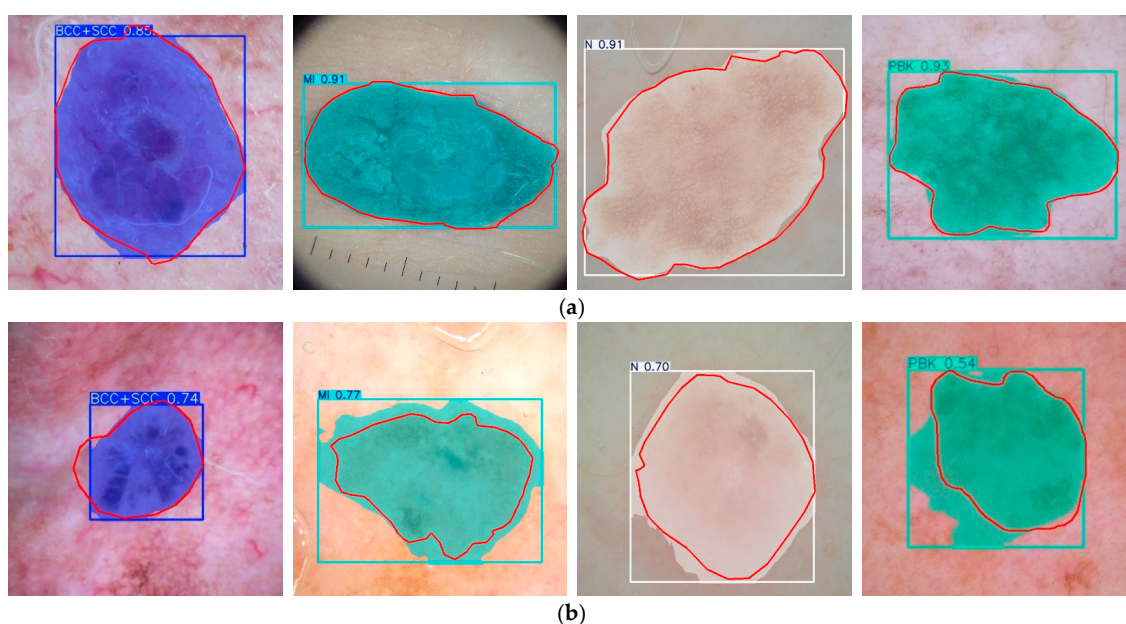


Figure 11. Example segmentation results returned by the yolov11s-seg model: (a) skin lesions identified with higher confidence; (b) skin lesions identified with lower confidence. The red line indicates the boundaries of the ground-truth mask. The images subjected to prediction had different resolutions. To facilitate observation of the results, they were cropped with an aspect ratio of 1:1.

5. Discussion

5.1. Comparison of Results with Those of Other Authors

The performance of our yolov11s-seg model has been presented in the context of a comparison with other relevant studies, which is summarized in Table 4. A direct comparison of the results is challenging due to significant differences in datasets, the number of classes, and the tasks performed (e.g., classification, segmentation, or the use of complex hybrid architectures). However, this broader analysis provides valuable insights into our model's performance within the current landscape of artificial intelligence research in dermatology.

Table 4. Comparison of results.

No. in Ref.	Model	Dataset	No. of Images	No. of Classes	Precision	Recall	F1-Score	mAP@50
Our model	yolov11s-seg	ISIC	7000	5	0.852	0.805	0.814	0.840
[38]	YOLOv4	ISIC	n/a	9	0.821	0.920	0.868	0.880
[39]	YOLOv11	Skin Disease Computer Vision Project	2486	10	0.920	0.980	0.950	0.920
[40]	YOLOv11	Kaggle	900	9	0.887	0.867	0.870	n/a
	ResNet	Kaggle	900	9	0.803	0.790	0.788	n/a
	VGG16	Kaggle	900	9	0.760	0.720	0.710	n/a
	YOLOv8	Kaggle	900	9	0.870	0.840	0.850	n/a
[31]	YOLOv8	ISIC 2020, HAM10000 and PH2	40,000	n/a	0.910	0.900	0.905	0.986
[30]	YOLOSAMIC	ISIC 2016–2018, private	4228	15	0.979	0.930	n/a	0.983
[41]	YOLOv10 + SAM	ISIC 2018	n/a	n/a	0.954	0.772	0.833	n/a
[42]	YOLOv9-M + Faster R-CNN with ResNet-50 core + FPN	ISIC 2019–2020 + HAM10000	68,472	n/a	0.951	0.944	0.951	0.971
[44]	YOLOv8 + ViT	HAM10000	10,015	7	0.742	0.717	0.729	0.870

Comparing our work with other studies, we notice that models trained on very large datasets or solving fewer complex problems often report higher metrics. For example, Albahli achieved outstanding mAP@50 results exceeding 0.98 using a YOLOv8 model trained on 40,000 images, and İnan et al. achieved a sensitivity of 0.98 using the YOLOv11 model. Similarly, Aishwarya et al. and Diptho and Basak reported high precision and sensitivity indices for the YOLOv4 (mAP of 0.88) and YOLOv11 architectures, respectively. The mAP@50 score of our model at 0.840 (with a sensitivity of 0.805 and a precision of 0.852) may seem slightly lower when juxtaposed with these record-breaking works; however, it should be emphasized that our task was significantly more complex. It involved the analysis of 5 classes of difficult skin lesions and an innovative background class (healthy skin), designed to minimize false positives.

To provide a more comprehensive comparison, Table 4 also includes studies focusing on computationally demanding hybrid architectures. Works such as those by Gül et al. (YOLOSAMIC) or Marie et al. (YOLOv9 combined with Faster R-CNN) demonstrate excellent mAP@50 scores in the range of 0.97–0.98. However, it should be noted that such approaches entail a drastic increase in computational complexity. Meanwhile, our baseline yolov11s-seg model exhibits a massive advantage in terms of inference speed (only 4.8 ms),

making it a practical and highly efficient tool for real-time mobile systems. Other hybrid approaches, such as integrating YOLOv10 with the SAM library (Nagaoka) or YOLOv8 with Vision Transformers (ViT) (AbuAlkebash et al.), achieved lower sensitivity values (0.772 and 0.717, respectively) compared to the 0.805 score obtained by our model.

In conclusion, although some studies report higher individual metrics, this often occurs in tasks of lower complexity (e.g., the absence of background images representing healthy skin in the training set) or when using heavy hybrid architectures. Our work establishes a new, more realistic benchmark in segmentation research. The results prove that the yolov11s-seg model is a robust and highly effective solution for the difficult task of skin lesion segmentation, especially considering the added complexity and the clinical necessity of correctly recognizing and ignoring healthy skin.

5.2. Comparison of the Best Model with Other Architectures

To extend the conducted research and obtain a broader comparative context, an additional experiment was carried out involving classical segmentation architectures available in the segmentation_models package [46]. As part of this stage, three models were built and trained: U-Net, LinkNet, and FPN. The selection of these architectures was motivated by their widespread use in semantic segmentation tasks and their diverse structures, which enable a more comprehensive analysis of the results.

To ensure comparability of results across models, the same neural network backbone—ResNet34—was used in all cases. Unifying the feature extraction layer made it possible to limit the impact of architectural differences to the decoding mechanisms and reconstruction of segmentation masks. The training process was conducted using identical hyperparameters for all models. The number of epochs was set to 50, the batch size to 8, and the input image resolution to 640 pixels. The learning rate was set to 0.0001, and a composite loss function, categorical_focal_dice_loss, was used, combining the properties of Dice loss and focal loss to improve robustness to imbalanced data.

During training, callback mechanisms were employed to support the optimization process. ModelCheckpoint was used to save the best model based on a selected validation criterion, and ReduceLROnPlateau dynamically reduced the learning rate when performance improvements stagnated. All models were trained and tested on the same dataset that had previously been used in experiments with the yolov11s-seg model. Additionally, the entire computational process was carried out on the same hardware configuration, ensuring consistency of experimental conditions and eliminating the influence of hardware differences on the results obtained.

After training, the resulting models were compared with the best solution identified in earlier research; namely, the yolov11s-seg model. The evaluation was conducted based on two key metrics: the F1 score, reflecting the balance between precision and recall, and inference time, which is important from the perspective of practical applications and system performance. The results obtained enabled a direct comparison between classical segmentation architectures and the YOLO-based approach, constituting a significant contribution to the analysis of the effectiveness of different segmentation methods in the problem considered.

The results of model testing are presented in Figure 12. There are clear differences between the models in terms of both accuracy, expressed by the F1-score metric, and execution speed. The yolov11s-seg model achieved the highest F1-score by a significant margin (81.4%). It clearly outperforms the other models in terms of accuracy—the difference exceeds 24 percentage points (FPN—56.6%, U-Net—55.0%, LinkNet—53.4%).

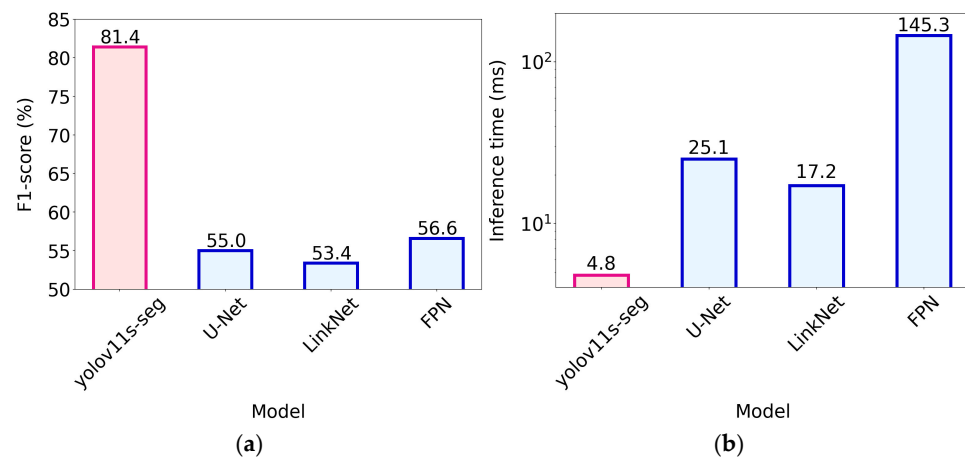


Figure 12. The results obtained for models from the segmentation_models package: (a) F1-score; (b) inference time.

In terms of execution speed, expressed as inference time, the yolov11s-seg model again proved to be the best. Its inference time is by far the shortest (4.8 ms) compared to the other models: LinkNet—17.2 ms, U-Net—25.1 ms, and FPN—145.3 ms (the slowest). Thus, the YOLOv11s-seg model proved to be not only the most accurate but also significantly the fastest. Therefore, if there are no additional constraints (e.g., hardware or architectural), this model can be recommended for practical applications—both in real-time systems and in scenarios where high inference precision is crucial.

It is also worth noting that many existing studies omit a background class, which simplifies the task to a closed-set classification or segmentation. By including healthy skin images, our model is forced to learn more discriminative features to avoid false positives, a common problem in real world clinical applications. While transformer-based model models like those can capture long range dependencies, they often require significantly more computational resources, which makes them less suitable for the real-time mobile deployment that our yolov11s-seg variant enables with its 4.8 ms inference time.

5.3. Limitations of the Study

Despite the high efficiency of the proposed system, several limitations must be acknowledged. First, the segmentation of basal cell and squamous cell carcinomas (BCC+SCC class) remains challenging due to their blurred boundaries and low contrast against healthy skin. Second, while the ISIC archive provides a vast variety of images, future work should include external validation on datasets with more skin phototypes to ensure global diagnostic equity. Finally, the current model operates solely on dermoscopic images—in this regard, integrating patient clinical data (e.g., age, history of sun exposure) from Electronic Medical Records (EMRs) could further enhance the specificity of the diagnostic process.

5.4. Practical Use of Research

Figure 13 presents the general architecture of the skin lesion segmentation system. The system consists of three logical layers: user, client, and server. At the highest level, interaction occurs through the user utilizing the client, which acts as the interface. The client is implemented as a mobile application in React Native v0.72.6 using TypeScript v5.2.2 and the Material UI v5.14.19 library. The client provides the following functions: image registration using a smartphone camera, preliminary image processing (cropping skin lesions, contrast enhancement, noise reduction, and resolution reduction), sending images to the server, and displaying the results returned by the neural network (prediction outcome) along with textual information for the user. Communication between the client

and the server takes place via the HTTP protocol using the axios library. The server component is implemented in the Flask framework (Python) and integrates the REST API interface module and the analytical module. The REST API module receives HTTP requests from the client, validates them, and forwards them to the server's analytical module. The main element of the analytical module is the yolov11s-seg model, which is used to predict skin lesions based on the input images sent to it. After executing the prediction, the REST API returns the results to the client in a standardized and formatted manner in JSON format. The prediction result is displayed to the user in the form of a skin lesion with a segmentation mask, the name of the detected skin lesion, the confidence of its detection, and advice regarding further steps.

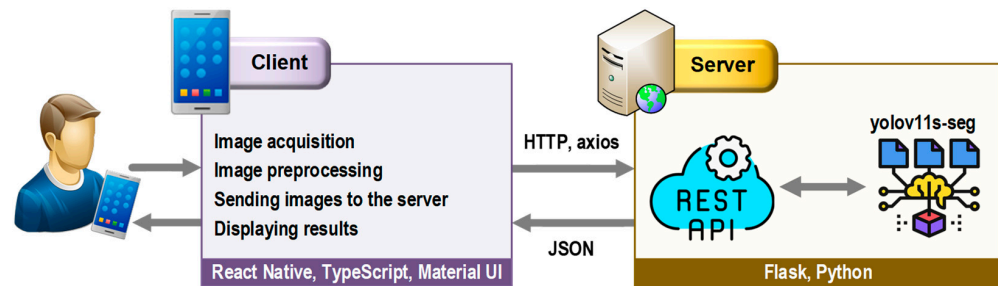


Figure 13. General architecture of the skin lesion segmentation system.

The proposed architecture has significant advantages, which include an instantaneous prediction process based on the highly efficient yolov11s-seg model. This design facilitates the use of a smartphone and a dedicated server for the immediate screening of patient lesions via the cloud, radically accelerating the preliminary diagnostic process. Furthermore, the network was trained to tolerate healthy skin, which minimizes the occurrence of false alarms—a crucial factor for the psychological well-being of patients. The main drawback of the solution remains its reduced sensitivity for poorly defined squamous cell and basal cell carcinomas (the BCC+SCC class) relative to medical expectations. Due to their immense visual similarity and blurred boundaries, standalone networks, without the application of powerful hybrid segmentation tools, will naturally experience difficulties with this specific case.

6. Conclusions

This study addresses the critical need for robust and reliable segmentation models in dermatological diagnosis. While the existing literature has explored various deep learning architectures, our work introduces several key novelties. The primary contribution of the research conducted is the creation and utilization of a new, balanced dataset. The skin lesion image dataset, based on the ISIC archive, was expanded with an additional background class containing images of healthy skin. Including negative samples is a crucial step that allows the model to effectively distinguish between the presence and absence of a lesion, drastically reducing the number of false alarms and ensuring diagnostic stability under realistic conditions.

We conducted a comprehensive analysis of the latest YOLO family architecture variants (v8, v9, v11, v12, v26) to identify the most promising solution for clinical decision support systems. Through a two-stage evaluation process, we rejected models of the highest complexity (such as yolov26x), which, despite high precision, were characterized by too long inference times for mobile devices. Ultimately, the yolov11s-seg model was selected as the superior architecture, exhibiting an optimal balance among high accuracy, precision, and computational efficiency, achieving an inference time of merely 4.8 ms. Most importantly, from the perspective of patient safety, this model achieved the highest sensitivity

(85.9%) in detecting invasive melanoma (MI)—the most aggressive and malignant of the examined lesions.

Despite its high overall effectiveness, the segmentation of lesions with blurred and indistinct edges, such as basal cell and squamous cell carcinomas (the BCC+SCC class), remains a challenge. There is a need for further research to improve model effectiveness, expand the dataset with images of lesions on darker skin tones, and integrate models with individual patient data from electronic medical records (EMRs). However, the achievements of this work prove that the optimized yolov11s-seg model constitutes a highly reliable solution ready for deployment in real-time teledermatology applications. To enable further innovation and collaboration within the scientific community, our prepared dataset has been made publicly available on the Zenodo platform, setting a new standard for future research on CAD systems in dermatology.

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